

SDLC, stakeholders and phases ★ Golden

One college period · 40–45 minutes · System Development Life Cycle

What you will learn

- Define system and SDLC
- Name stakeholder roles
- Describe planning, analysis, design, coding, testing, deployment, maintenance

System. A set of related components that work together toward a goal. SDLC is the planned process of building information software from idea to retirement.

Stakeholder	Role
Project manager	Plan, cost, schedule
Business analyst	Gather requirements
System architect	Overall design
Developer	Write code
Tester	Find bugs
Operations	Deploy on servers
Production support	Maintenance after go-live

Planning → Requirement analysis → Design → Coding → Testing → Deployment →

Maintenance

- Planning: problem, feasibility (technical, economic, operational), schedule.
- Requirement engineering: functional (what it must do) vs non-functional (speed, security). SRS document.
- Design: architecture, algorithms, flowcharts, database schema, UI.
- Implementation: coding in C / VB / other.
- Testing: find bugs and errors; unit, integration, system, acceptance.
- Deployment & SLA: install, train, service-level agreement.
- Maintenance: fix, adapt, improve.

Models you may mention: Waterfall (linear, old board favourite) vs Agile (iterative sprints).

اردو میں سمجھیں

SDLC سافٹ ویئر بنانے کا منصوبہ بند چکر ہے: منصوبہ، ضرورت، ڈیزائن، کوڈ، ٹیسٹ، تنصیب، دیکھ بھال۔

CHECK YOURSELF

1. SRS means?
2. Who gathers requirements?

Answers: 1. Software Requirements Specification · 2. Business analyst

MCQs

1. SDLC is mainly about:
 - a. repairing printers
 - b. developing software in stages
 - c. only testing viruses
 - d. drawing cables
2. Converting design into programs is:
 - a. testing
 - b. coding / implementation
 - c. feasibility
 - d. retirement

Key: 1=b, 2=b

Short questions (3 marks each)

1. Define SDLC and list its phases.
2. Differentiate functional and non-functional requirements.
3. Who are the stakeholders of SDLC?

Model short answers

1. Life cycle of software: plan, analyse, design, code, test, deploy, maintain.
2. Functional: features (print marksheet). Non-functional: quality (response in 2 seconds, password security).
3. PM, analyst, architect, developers, testers, operations, support, plus the client/users.

Long questions (board Section C)

1. Explain the System Development Life Cycle with a diagram and the role of each stakeholder.